INTRODUCTION TO INFORMATION GEOMETRY

Natural Gradients and Replicator Flows

Week 9 · Lecture 1

Christopher Lee

Which direction increases a function fastest?

For a function $\Phi : \mathbb{R}^n \to \mathbb{R}$, the ordinary gradient is defined by

$$d\Phi_p(v) = \nabla\Phi(p) \cdot v.$$

This dot product measures length using Euclidean geometry. Probability distributions come with a different way to measure the length of a small change.

A gradient depends on the metric

Given a metric g, define the *metric gradient* by

$$g_p(\operatorname{grad}_g \Phi, v) = d\Phi_p(v) \quad \text{for every tangent vector } v.$$

In local coordinates, write $g(v, w) = v^T G(p) w$. Then

$$\operatorname{grad}_g \Phi = G(p)^{-1} \nabla \Phi.$$

Fisher geometry on a Bernoulli family

For $P = (x, 1-x)$, the Fisher line element is

$$ds_F^2 = \frac{dx^2}{x(1-x)}.$$

Its metric coefficient is $G(x) = 1/[x(1-x)]$. Consequently the Fisher-gradient ascent equation is

$$x' = x(1-x)\Phi'(x).$$

An ascent flow on the Bernoulli interval

Take the potential $\Phi(x) = 3x - 2x^2$. Find its natural-gradient equation and any interior equilibrium.

$$x' = x(1-x)\Phi'(x).$$

An ascent flow on the Bernoulli interval

Take the potential $\Phi(x) = 3x - 2x^2$. Since $\Phi'(x) = 3 - 4x$,

$$x' = x(1-x)(3-4x).$$

The interior equilibrium is $x_* = 3/4$.

The phase line of this flow



The potential rises until $x = 3/4$ and falls after it. The gradient flow moves toward the interior maximum.

Why the potential increases

Along a solution $x(t)$, the chain rule gives

$$\frac{d\Phi}{dt} = \Phi'(x)x'.$$

Substitute the Fisher-gradient equation:

$$\frac{d\Phi}{dt} = x(1-x)[\Phi'(x)]^2 \geq 0.$$

The identity expresses ascent without requiring us to solve the ODE.

How fast is the potential increasing?

At $x = 1/2$, for $\Phi(x) = 3x - 2x^2$, calculate

$$\frac{d\Phi}{dt} = x(1-x)(3-4x)^2.$$

Also find the maximal value of Φ on $[0, 1]$.

How fast is the potential increasing?

At $x = 1/2$, we have $3 - 4x = 1$, so

$$\frac{d\Phi}{dt} = \tfrac{1}{2} \cdot \tfrac{1}{2} \cdot 1^2 = \tfrac{1}{4}.$$

The maximum occurs at $x = 3/4$:

$$\Phi(3/4) = \tfrac{9}{8}.$$

Tangent vectors on a simplex

For $p = (p_1, \dots, p_n)$, the coordinates satisfy

$$\sum_i p_i = 1.$$

Every tangent vector v therefore satisfies $\sum_i v_i = 0$. The categorical Fisher metric is

$$g_p(v, w) = \sum_i \frac{v_i w_i}{p_i}.$$

The Fisher gradient on the simplex

Let $a_i = \partial\Phi/\partial p_i$, and set $\bar{a} = \sum_j p_j a_j$. The Fisher gradient has coordinates

$$(\mathrm{grad}_F\,\Phi)_i = p_i(a_i - \bar{a}).$$

Its coordinates sum to zero, so the resulting velocity is tangent to the probability simplex.

A gradient at the center of the simplex

Let $\Phi(p) = 2p_A + p_B$, and take $P = (1/3, 1/3, 1/3)$. Find $\bar{a}$ and the Fisher-gradient velocity

$$p_i' = p_i(a_i - \bar{a}), \quad a = (2, 1, 0).$$

A gradient at the center of the simplex

Let $\Phi(p) = 2p_A + p_B$, and take $P = (1/3, 1/3, 1/3)$. The weighted mean is $\bar{a} = (2 + 1 + 0)/3 = 1$. Hence

$$p' = (1/3,\ 0,\ -1/3).$$

Type A increases, type B is momentarily constant, and type C decreases.

Replicator dynamics is a Fisher gradient

If the payoff or fitness values have the form

$$f_i(p) = \frac{\partial \Phi}{\partial p_i},$$

then the natural-gradient flow becomes

$$p_i' = p_i(f_i - \bar{f}).$$

This is exactly the replicator equation, now understood as motion up a potential in Fisher geometry.

Variance measures both speed and ascent

For the flow $p' = \operatorname{grad}_F \Phi$,

$$\frac{d\Phi}{dt} = d\Phi_p(p') = g_p(p', p').$$

Using $p_i' = p_i(f_i - \bar{f})$ gives

$$\frac{d\Phi}{dt} = \|p'\|_F^2 = \operatorname{Var}_p(f).$$

Fitness variance is the squared geometric speed of ascent.

Check the ascent identity numerically

For $P = (1/3, 1/3, 1/3)$ and $\Phi = 2p_A + p_B$, calculate the instantaneous increase in Φ. Use $p' = (1/3, 0, -1/3)$ and $\dot{\Phi} = 2p'_A + p'_B$.

Check the ascent identity numerically

For $P = (1/3, 1/3, 1/3)$ and $\Phi = 2p_A + p_B$,

$$\dot{\Phi} = 2(1/3) + 0 = 2/3.$$

The fitness variance gives the same answer:

$$\tfrac{1}{3}(2-1)^2 + \tfrac{1}{3}(1-1)^2 + \tfrac{1}{3}(0-1)^2 = \tfrac{2}{3}.$$

The boundary and the metric

The formula $p_i' = p_i(f_i - \bar{f})$ makes each face $p_i = 0$ invariant. The Fisher metric becomes singular in these coordinates at that face. Interior trajectories remain in the positive simplex at every finite time when the fitness functions are sufficiently regular.

A question for the next lecture

We have described motion that increases a chosen potential. Which potential should we choose if we want a population of probabilities to approach a specified target distribution? Relative entropy supplies two instructive answers.